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RESEARCH ARTICLE

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Regio- and Stereospecific Copper-Catalyzed S_N2' Substitution Reaction of Propargyl Esters with Polyfluoroaryl Zinc Reagents

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Dedication to Professor Armido Studer for receiving the Paracelsus Prize 2024 of the Swiss Chemical Society

Copper-catalyzed S_N2' substitution reactions of chiral propargyl compounds provides an efficient synthetic method for chiral allenes that have widespread applications in pharmaceuticals and artificial functional materials. However, compared with the S_N2' reaction catalyzed by palladium, ruthenium, and other noble metals, the development of S_N2' reaction catalyzed by copper is relatively delayed due to its sensitivity to steric hindrance. Herein we reported a highly selective copper-catalyzed S_N2' substitution reaction through the use of weakly nucleophilic zinc reagents. A wide variety of polyfluoroaryl allenes bearing an assortment of functional groups (36 examples) are compatible with this catalysis system. It is worth noting that chiral polyfluoroaryl allenes can also be efficiently constructed from chiral propargyl esters via this S_N2' reaction. A series of mechanism verification experiments and DFT calculations also provide a basis for the reaction mechanism and the stereo-model of chiral transfer. Moreover, the preparation of other valuable polyfluoroarene containing compounds from polyfluoroaryl allenes also proves the synthetic utility of this methodology.

Keywords: Chiral allenes, Copper-catalysis, Fluorochemicals, Regio and stereospecific, S_N2' reaction

1. Introduction

Axially chiral allenes are a class of dominant skeletons, which exist in many natural products, drug molecules, ligands, and molecular structures of advanced functional materials.[1–3] Since Burton and Pechmann's pioneering work in 1887,[4] the development of synthetic strategies for efficient construction of chiral allenes has long been a research hotspot in organic synthesis.[5–9] Among the many developed methods, transition metal-catalyzed S_N2' substitution reactions of chiral propargyl electrophiles with nucleophiles is favored by synthetic chemists because of the avoidance of complex chiral ligands and the ease of preparation of chiral propargyl electrophiles.[10] At present, the S_N2' substitution reactions catalyzed by palladium, rhodium, and other noble metals have been relatively mature.[11–18] However, compared with the former, the copper-catalyzed S_N2' reaction is relatively delayed, mainly because the regioselectivity of the reaction decreases as the steric hindrance of the nucleophile or substrate increases.[19–23] As a drawback, copper-catalyzed substitution reaction of propargyl ester has two possible reactive sites (α and γ positions) that could lead to mixtures of propargyl or allenyl products (Figure 1a).[24–26] Therefore, achieving the synthesis of chiral allenes through highly selective copper-catalyzed S_N2' reaction is still very significant and challenging.

Polyfluoroarenes widely exist in artificial functional materials (i.e., pharmaceuticals, materials) due to their inherent ability to form special intermolecular interactions.[27–31] In addition, the polyfluoroaryl groups can also form Z-type interactions with transition metals when incorporated into ligands, thereby

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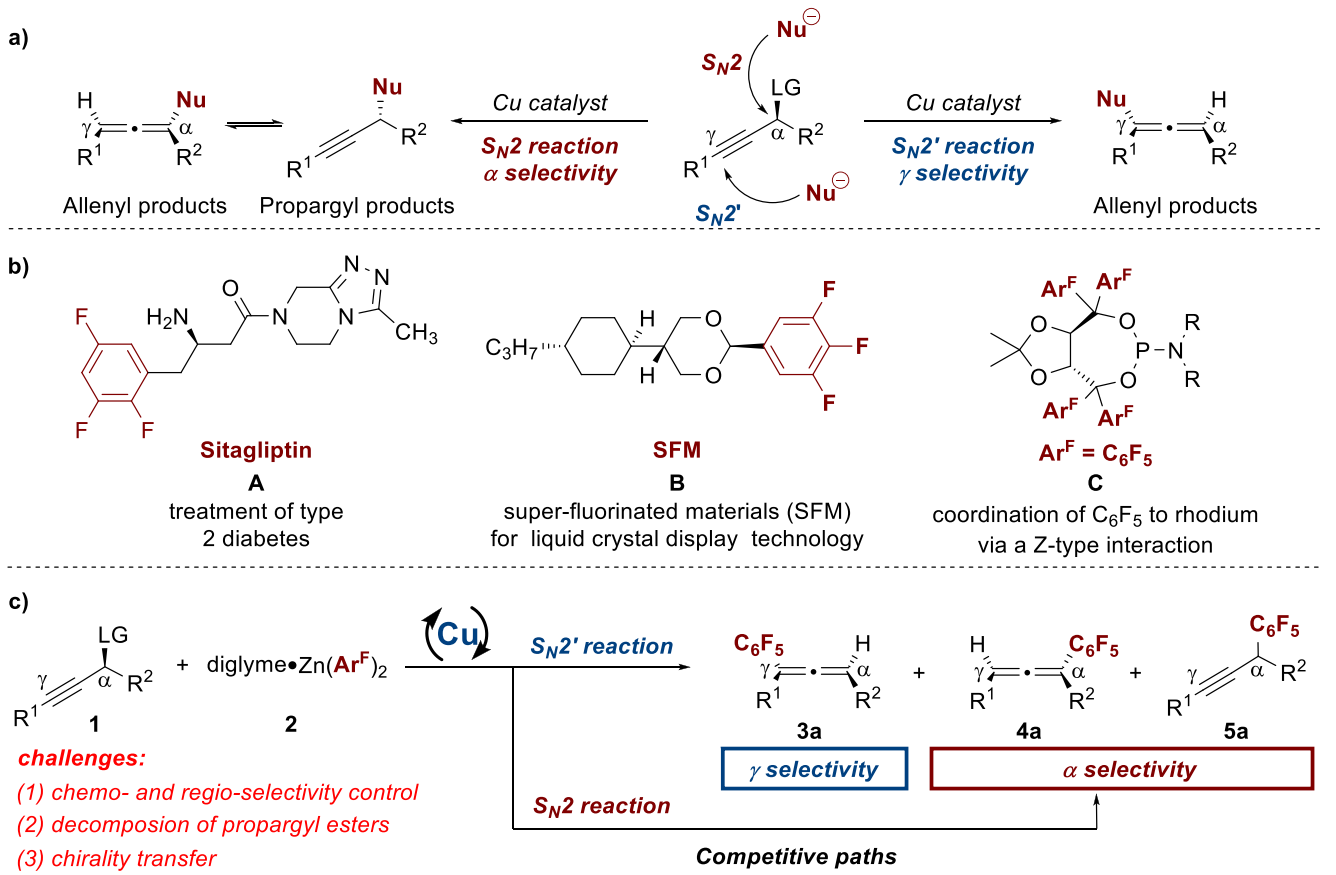


Figure 1. Background of the copper-catalyzed substitution reaction of propargyl ester and the artificial functional materials containing polyfluoroarenes. a) Copper-catalyzed S_N2 and S_N2' substitution reaction of propargyl ester. b) Artificial functional materials containing polyfluoroarenes. c) This work: regio- and stereospecific copper-catalyzed S_N2' substitution reaction of propargyl esters with polyfluoroaryl zinc reagents

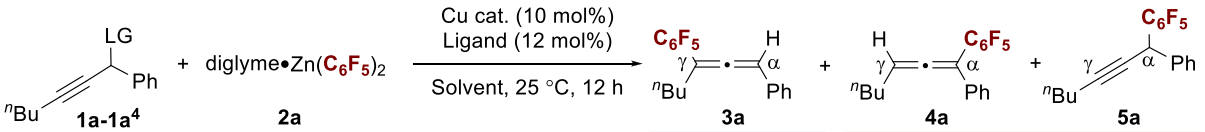
improving the selectivity of the reaction (Figure 1b).[32,33] Precisely because of these important applications, a number of protocols have been developed for synthesizing polyfluoroaryl-containing compounds. However, copper-catalyzed methods to synthesize chiral polyfluoroaryl allenes are still underdeveloped. To our knowledge, the regioselectivity and stereospecificity of the reaction have not yet been well controlled due to the requirement rigid ligand and strong base.[23] It illustrates the challenge of achieving chiral polyfluoroaryl allenes with high selectivity via S_N2' reactions. Herein we report the first regio- and stereospecific method for the preparation of chiral polyfluoroaryl allenes from readily available chiral propargyl esters[34,35] and polyfluoroaryl zinc reagents[36–39] through copper-catalyzed S_N2' substitution reactions (Figure 1c). This catalysis system successfully suppresses the competing S_N2 side reaction and overcomes the problem of regioselectivity and stereospecificity.

2. Results and Discussion

We began to test our hypothesis using the racemic propargyl ester **1a** as the propargyl electrophiles and polyfluoroaryl zinc reagent **2a** instead of the traditional combination of polyfluorobenzene and a strong base which could avoid the racemization of propargyl ester. To regulate the regioselectivity, a variety of ligands including bipyridine, phalenoline and pyridinoxazoline were employed firstly (Scheme 1, entries 1∼5, for the details of the investigation of ligands, see Supporting Information). A curious thing is that in the presence of these ligands, the allene product **3a** can be obtained with almost perfect γ-selectivity (Scheme 1, entry 6). We hypothesize that diglyme (diethylene glycol dimethyl ether), which plays a stabilizing role in zinc reagents, may act as the ligand in the reaction. The results of copper catalyst evaluation (Scheme 1, entries 7∼9) showed that when $Cu(OTf)_2$ is used instead of $Cu(MeCN)_4BF_4$, the reaction can give the product with good yield and good

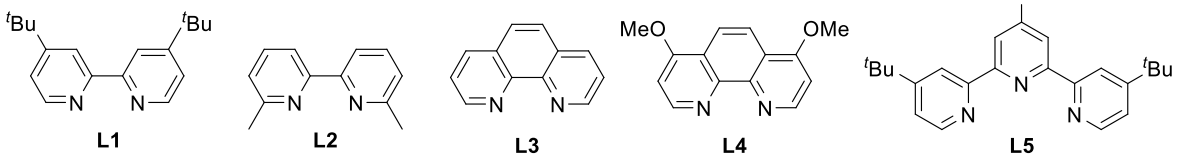
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for entries 1-12: LG = 3,5(CF$_3$)$_2$-BzO (**1a**); for entry 13: LG = AcO (**1a^1**), BzO (**1a^2**), MeOCO$_2$ (**1a^3**), (EtO)$_2$PO$_2$ (**1a^4**)

entry	sub. **1**	Cu cat.	ligand	solvent	yield of **3a** + **4a** + **5a** [%][a]	γ/α[b]
1	**1a**	Cu(MeCN)$_4$BF$_4$	**L1**	DCM	44	> 99 : 1
2	**1a**	Cu(MeCN)$_4$BF$_4$	**L2**	DCM	51	> 99 : 1
3	**1a**	Cu(MeCN)$_4$BF$_4$	**L3**	DCM	86	> 99 : 1
4	**1a**	Cu(MeCN)$_4$BF$_4$	**L4**	DCM	62	> 99 : 1
5	**1a**	Cu(MeCN)$_4$BF$_4$	**L5**	DCM	64	> 99 : 1
6	**1a**	Cu(MeCN)$_4$BF$_4$	-	DCM	87	> 99 : 1
7	**1a**	CuBr	-	DCM	77	> 99 : 1
8	**1a**	Cu(OAc)$_2$	-	DCM	89	> 99 : 1
9	**1a**	Cu(OTf)$_2$	-	DCM	99 (90)[c]	> 99 : 1
10	**1a**	Cu(OTf)$_2$	-	THF	99	> 99 : 1
11	**1a**	Cu(OTf)$_2$	-	EtOAc	99	> 99 : 1
12	**1a**	Cu(OTf)$_2$	-	MeCN or DMF	NR	-
13	**1a^1 - 1a^4**	Cu(OTf)$_2$	-	DCM	0-62	> 99 : 1



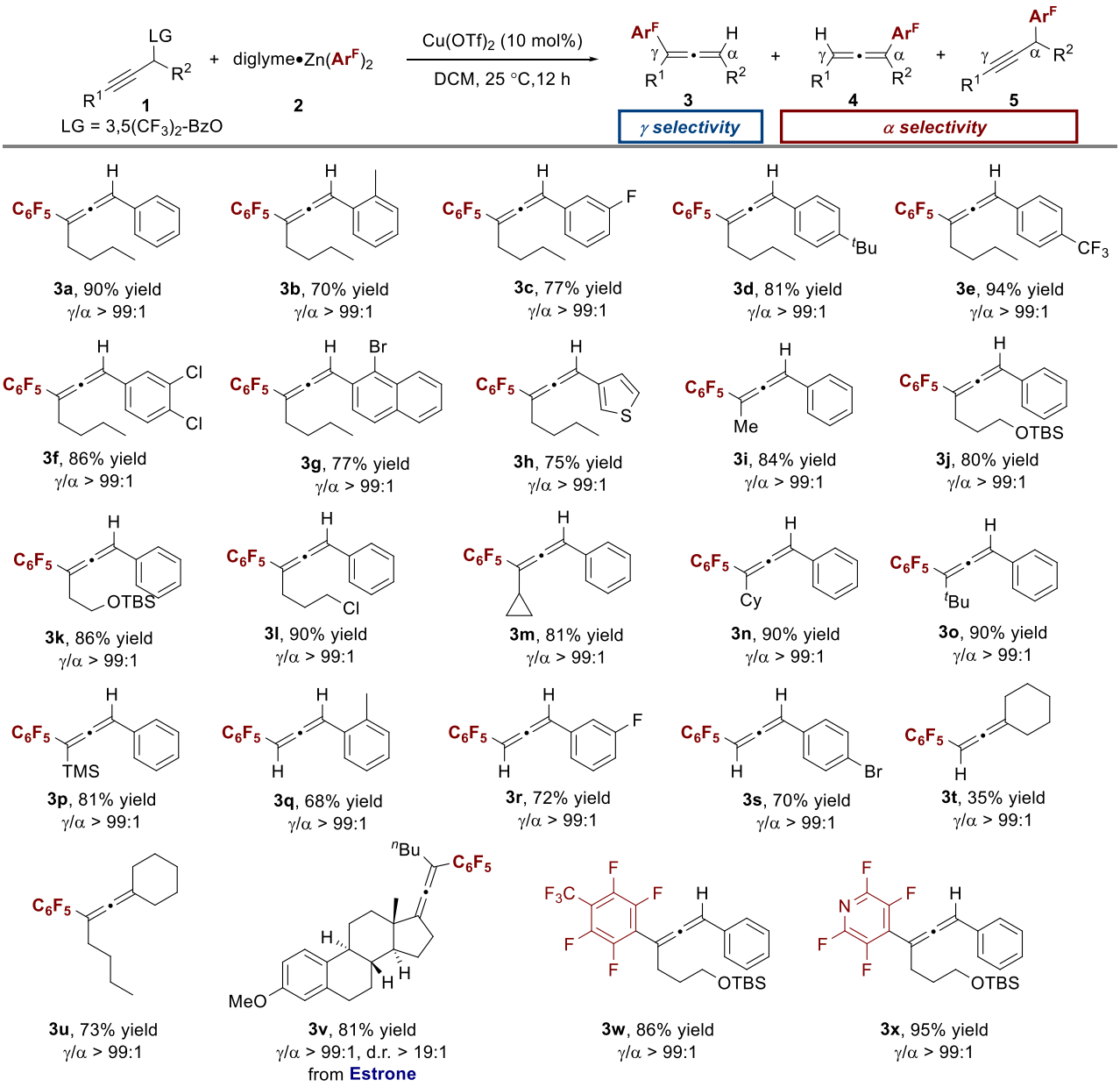
Scheme 1. Investigation of reaction conditions. Reactions were run on 0.2 mmol scale in 2 mL of dry DCM for 12 h under Ar. [a]The yield was determined by ^{1}H NMR spectroscopy with 1,3,5-trimethoxybenzene as an internal standard. [b]The ratio of γ/α was confirmed by ^{1}H NMR spectroscopy. [c]Yield of isolated product is given in the parentheses. [d]Ac, acetyl; Bz, benzoyl.

regioselectivity (99% yield, >99:1 γ/α). No better result was given when DCM was instead by other solvents (Scheme 1, entries 10∼12). The use of other propargyl esters bearing other leaving groups (e.g., OAc, OBz, OCO$_2$Me, etc.) led to a decrease of yield to varying degrees (Scheme 1, entry 13).

With the optimized conditions in hand, we next prepared a variety of racemic propargyl esters, and first evaluated the structural scope. As shown in Scheme 2, a variety of propargyl esters and polyfluoroaryl zinc reagents all smoothly participated in this S_N2' reaction with high γ/α selectivity. The variation of electrical properties and positions of the substituents on the benzene ring do not affect the yield and γ/α selectivity of the reaction (**3a∼3f**, 70%–94% yield, up to >99:1 γ/α). Moreover, propargyl esters bearing naphthyl and heteroaryl substituents were also tolerated by this catalytic system (**3g**, 77% yield; **3h**, 75% yield, up to >99:1 γ/α). Primary alkyl substituted propargyl esters can be very compatible in the reaction, especially containing OTBS, Cl groups (**3i∼3l**, 80%–90% yield, up to >99:1 γ/α). To our delight, the regioselectivity of this reaction is less affected by steric hindrance, which is different from the previously reported copper-catalyzed S_N2' reactions. Both the sterically demanding secondary and tertiary alkyl substitutes can be applied in this reaction smoothly without decreasing regioselectivity (**3m∼3p**, 81%–90% yield, up to >99:1 γ/α). In addition to trisubstituted allenes, 1, 3-disubstituted allenes can also be successfully obtained under this copper catalytic strategy (**3q∼3s**,

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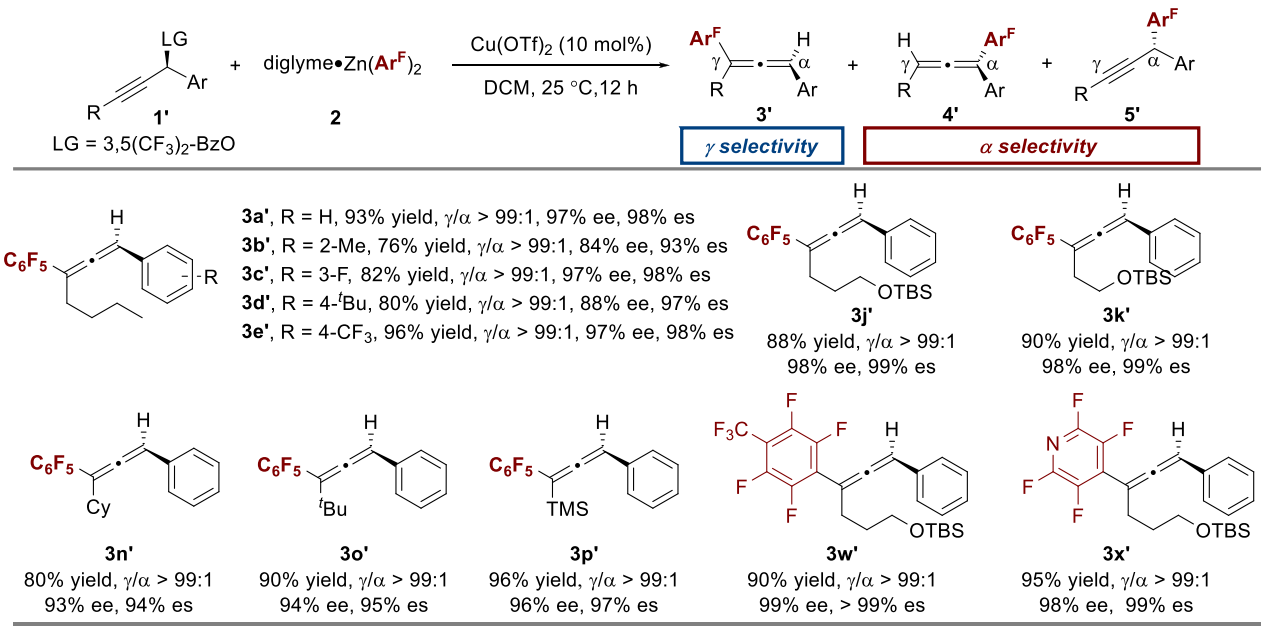
Scheme 2. Scope for the synthesis of racemic polyfluoroaryl allenes. [a] All the reaction were performed in 0.2 mmol scale; isolated yields; ratios of γ/α were determined by 1H NMR spectroscopy. TBS: tertbutyl dimethyl silicyl.

68%–72% yield, up to >99:1 γ/α). Besides, the highly selective conversion of fully alkyl substituted propargyl esters, especially **1v** derived from Estrone, also proves the widespread substrate scope (**3t~3v**, 35%–81% yield, up to >99:1 γ/α).[40] To our delight, zinc reagents containing trifluoromethyl and pyridine, which have never been reported before, can also be efficiently converted to allenes (**3w**, 86% yield; **3x**, 95% yield, up to >99:1 γ/α).

Next, we checked whether the stereospecific S_N2' reaction of the chiral propargyl esters is as general as the racemic version. As shown in Scheme 3, like racemic propargyl esters, various representatively chiral propargyl esters were proven successful for this stereospecific transformation. Irrespective of the position and electrical properties of the substituents on the benzene ring, the corresponding chiral polyfluoroaryl allenes were efficiently afforded in good γ/α selectivity and with excellent chirality transfer (**3a'~3e'**, 76%–96% yield, 93%–98% es, up to >99:1 γ/α). When the steric hindrance of the alkyl chain on the propargyl ester is

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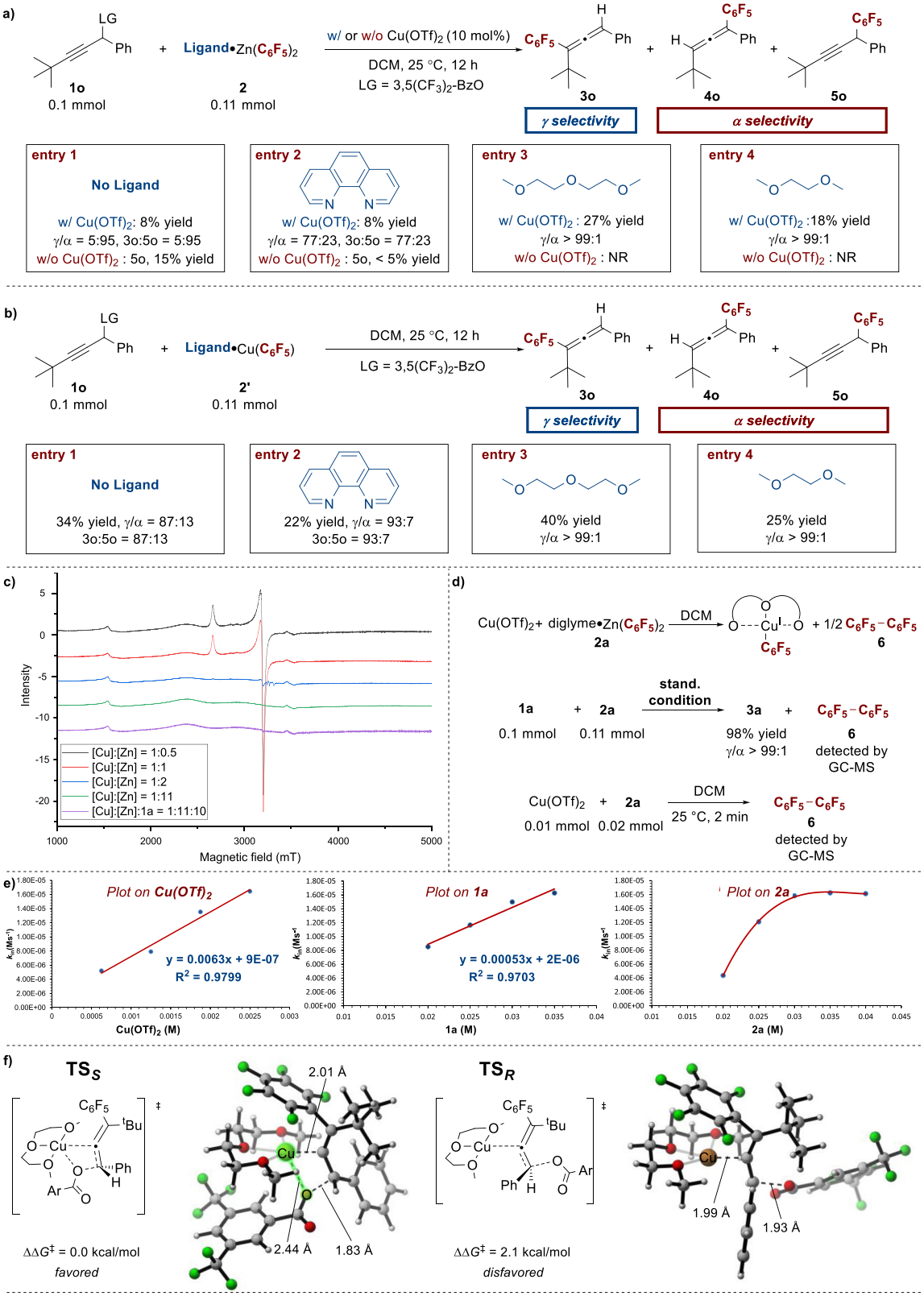
Scheme 3. Scope for the synthesis of chiral polyfluoroaryl allenes. [a] All the reaction were performed in 0.2 mmol scale; isolated yields; ratios of γ/α were determined by ^{1}H NMR spectroscopy; ee values were determined by chiral HPLC analysis TBS: tertbutyl dimethyl silicyl.

gradually increased, the reaction can also occur smoothly with high γ/α selectivity and high enantioselectivity maintenance (**3j'~3p'**, 80%–96% yield, 94%–99% es, up to >99:1 γ/α). Besides pentafluorophenyl, zinc reagents containing trifluoromethyl and pyridine are also suitable for this stereospecific copper catalytic system (**3w'~3x'**, 90%–95% yield, up to >99% es, > 99:1 γ/α). The es values of most of the reactions are above 95%, which proves the high stereospecificity of the reactions.

A series of experiments were conducted to shed light on the reaction mechanism. First, in order to further verify whether ligands play a decisive role in the reaction selectivity, we investigated the effect of zinc reagents chelated with different ligands. The γ/α selectivity of the reaction was decreased with either no ligand or with the highly steric 1,10-phenanthroline as the ligand, indicating that the ligand played a decisive role in the selectivity control (Scheme 4a, entries 1–3). We also found that the γ/α selectivity maintained when 1, 2-dimethoxyethane replaced diglyme as a ligand (Scheme 4a, entries 4). These results indicate that diglyme is most likely coordinated as a didentate ligand in the γ/α-selective determination step. Subsequently, we investigated the reaction of equivalent polyfluoroaryl copper with propargyl ester **1o**, and found that only the polyfluoroaryl copper reagent coordinated with diglyme/1,2-dimethoxyethane can give a good γ/α selectivity (Scheme 4b). Based on the above outcomes and literature information, we believe that the excellent selectivity of the reaction is due to the ability of the tridentate diglyme to form a relatively stable complex with copper and zinc reagents, thus weakening their nucleophilicity and avoiding the S_N2 reaction. In order to further prove the valence state of the copper catalyst in the reaction, we also conducted a series of EPR experiments (Scheme 4c). During the process of reaction, we did not observe the Cu(II) signal, which indicates that there may be no Cu(II) intermediates existed in the system. We also independently monitored the copper catalyst and zinc reagent via EPR, and found that the Cu(II) signal disappeared as the ratio of zinc reagent to $Cu(OTf)_2$ increased. Based on the above results, we suspect that the Cu(II) in the reaction may be reduced to Cu(I) by zinc reagent, along with the formation of fully fluorinated biphenyl **6** as a byproduct (Scheme 4d, up).[41,42] Indeed, we did observe the formation of byproduct **6** in the reaction system. The control experiments show that this byproduct can be generated when $Cu(OTf)_2$ reacts with zinc reagent, which proves our conjecture that zinc reagent reduces Cu(II) to Cu(I). In addition, in order to further understand the reaction mechanism, the kinetic studies were also carried out. The results showed first-order kinetic dependence on $Cu(OTf)_2$ and propargyl ester **1a**, and showed second-order kinetic dependence on the zinc reagent **2a** (Scheme 4e). Finally, DFT calculations were further carried out for the β-O elimination step to probe the origin of the chirality transfer. The difference in energy between two stereo-determining

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Scheme 4. Mechanistic experiments. a), Reactions of **1o** and different zinc reagents. b), EPR experiments with $Cu(OTf)_2$, diglymeZn(C_6F_5) and **1a**, EPR spectra of Cu(II) (X band, $T = 100$ K, $F = 9.394061$ GHz, $g\perp = 2.51$, $g\| = 2.10$). **c**, Byproduct determination. d), Kinetic studies. e), DFT calculation.

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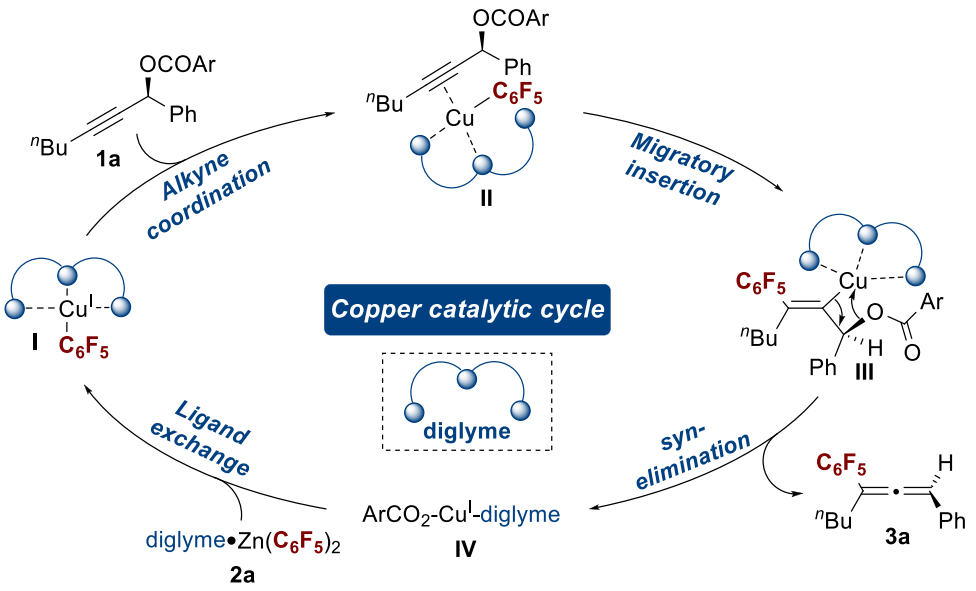


Figure 2. Proposed mechanism.

transition states was calculated to be 2.1 kcal/mol, in good accordance with the experimentally observed excellent selectivity (Scheme 4f). During the elimination process, **TS$_S$**, the favored transition state, witness a four-membered ring in which Cu facilitates the dissociation of carboxylate group through electrostatic interaction (distance of Cu–O is 2.44 Å, highlighted in Scheme 4f). On the other hand, the corresponding Cu–O interaction is absent in the disfavored transition state **TS$_R$** since the carboxylate group and Cu sit on the opposite sides of allene. The calculated natural population analysis (NPA) charge and non-covalent interaction analysis (NCI plot) also supported the existence of this electrostatic interaction (see Figure S8 in supporting information) in the more favored transition state **TS$_S$**, which rationalizes the high efficiency of chirality transfer observed by experiments.

On the basis of the above mechanistic studies and related literature, a possible catalytic cycle was proposed in Figure 2. First, the coordination and migratory insertion between the C_6F_5-Cu(I)-diglyme **I** and alkyne occur to delivered the intermediate **III**. Then the intermediate **III** would undergo rapid syn-elimination to form chiral allene products, accompanied by the formation of Cu(I) carboxylate **IV**. Finally, the Cu(I) carboxylate species and aryl zinc reagents underwent ligand exchange to regenerate Cu (I), completing the catalytic cycle.

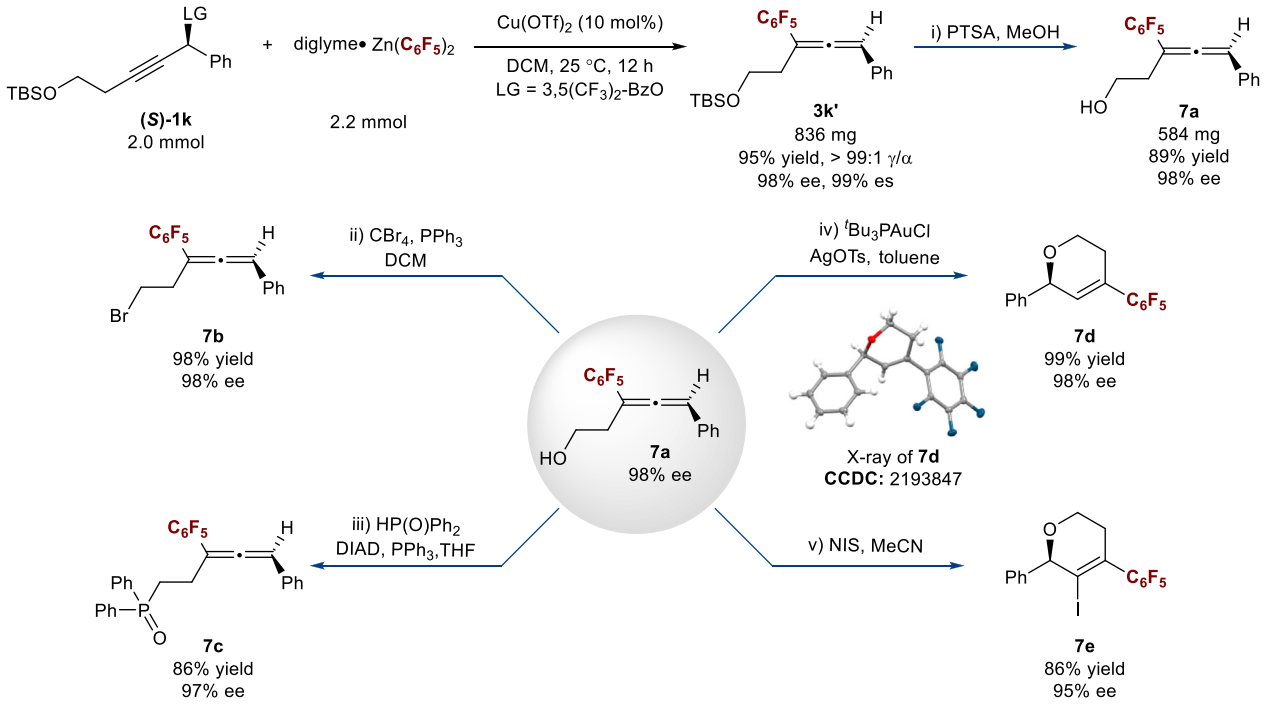
To demonstrate the synthetic utility of this S_N2' reaction, a scale-up reaction was performed to obtain the OTBS substituted chiral polyfluorinated arylene (*S*)-**3k'** in 95% yield and with 98% ee and >99:1 γ/α selectivity (in Scheme 5). In the presence of PTSA, the TBS group in **3k'** can be efficiently removed and converted into the hydroxy-substituted allene product **7a** in 89% yield with 98% ee.[43] Under the conditions of appeal reaction and Mitisunobu reaction, **7a** was converted to allenyl bromide **7b**[44] in 98% yield with 98% ee and allenyl phosphate **7c**[45] in 86% yield with 97% ee, respectively. Besides, **7a** can also undergo gold-catalysed cyclization to obtain axis-to-center chiral dihydropyran **7d** in 99% yield without the loss of enantioselectivity.[46] The absolute configuration of **7d** has been confirmed by single crystal X-ray diffraction. Additionally, **7a** was also readily cyclized with NIS to form chiral dihydropyran **7e** in 86% yield and with 98% ee.[47]

3. Conclusions

In summary, we have developed a regio- and stereospecific copper-catalyzed S_N2' substitution reaction for synthesizing structurally diverse chiral polyfluoroaryl allenes. The key to success is the use of weakly nucleophilic Zn-Ar^F reagents with weak nucleophile as the nucleophile and diglyme with low steric hindrance as ligand. Moreover, the synthetic utility of this methodology for the preparation of other valuable polyfluoroarene containing compounds has also been demonstrated. Importantly, we proposed a detailed mechanism and stereo-control model for the reaction based on the experimental and computational evidences, which provided a valuable reference for the subsequent study of

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Scheme 5. Synthetic application. [a] Reaction conditions: (i) PTSA (10 mol%), MeOH, 25 °C, 0.5 h. (ii) CBr_4 (1.3 equiv.), PPh_3 (1.5 equiv.), DCM, 0 °C, 12 h. (iii) $HP(O)Ph_2$ (1.0 equiv.), PPh_3 (1.1 equiv.), DIAD (1.2 equiv.), THF, 25 °C, 24 h. (iv) $^tBuPAuCl$ (10 mol%), AgOTs (10 mol%), toluene, 25 °C, 8 h. (v) NIS (3.0 equiv.), MeCN, -10 °C, 48 h. PTSA: p-Toluenesulfonic Acid, DIAD: Diisopropyl azodicarboxylate, NIS: *N*-Iodosuccinimide.

this kind of S_N2' reaction. We anticipate that the present approach will provides a new avenue for the synthesis of chiral polyfluoroaryl allenes and may have potential applications in medicinal and synthetic chemistry.

4. Experimental Section

4.1. Generic Procedure for Copper-Catalyzed S_N2' Substitution Reaction

In an argon-filled glove box, a flame-dried 10 mL Schlenk tube equipped with a magnetic stirrer bar and rubber plug was charged sequentially with $Cu(OTf)_2$ (7.23 mg, 0.02 mmol), zinc reagents **2** (117.4 mg, 0.22 mmol), and propargyl esters **1** (85.6 mg, 0.2 mmol), followed by the addition of DCM (2.0 mL). The vial was closed and the Schlenk tube was removed from the glove box. The mixture was then stirred at room temperature for 12 h. The product was purified by flash column chromatography on silica gel (PE/EA 100:0) to afford the desired product.

4.2. Generic Procedure for Removement of TBS Group

To a flame-dried bottle, **3k'** (881.6 mg, 2.0 mmol, 1.0 equiv., 98% ee) was dissolved in MeOH (12 mL). The mixture was added PTSA under stirring until the pH = 4 and then stirred at room temperature for 2 h. After that, the mixture was neutralized with NEt_3 and purified by flash column chromatography on silica gel (PE/EA 5:1) to afford **7a** (583.9 mg, 89% yield, 98% ee) as colorless oil.

4.3. Generic Procedure for Halogenation of Hydroxyl Group

To a flame-dried 10 mL Schlenk tube, **7a** (65.2 mg, 0.2 mmol, 1.0 equiv., 98% ee) was dissolved in DCM (2 mL). The mixture was cooled to 0 °C and stirred for 10 min. To the resulting solution was added CBr_4 and PPh_3, and the mixture was stirred for 4 h at 0 °C. After completion of the reaction (TLC), the resulting mixture was purified by flash column chromatography on silica gel (PE/EA 100:0) to afford **7b** (76.1 mg, 98% yield, 98% ee) as colorless oil.

4.4. Generic Procedure for Phosphorization of Hydroxyl Group

To a flame-dried 10 mL sealing tube, **7a** (65.2 mg, 0.2 mmol, 1.0 equiv., 98% ee), diphenylphosphine oxide (44.5 mg, 0.22 mmol, 1.1 equiv.) and PPh_3 (62.9 mg, 0.24 mmol, 1.2 equiv.) was dissolved in THF (1.0 mL). The mixture was cooled to 0 °C and di-isopropyl azodicar-

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boxylate was then added slowly. The reaction system was warmed to rt gradually and stirred for overnight. After completion of the reaction (TLC), the resulting mixture was purified by flash column chromatography on silica gel (PE/EA 2:1) to afford **7c** (87.5 mg, 86% yield, 97% ee) as colorless oil.

4.5. Generic Procedure for Au Catalysed Cyclization Reaction

To a flame-dried 10 mL sealing tube, tBu_3PAuCl (8.7 mg, 0.02 mmol, 0.1 equiv.) and AgOTs (5.6 mg, 0.02 mmol, 0.1 equiv.) was dissolved in toluene (1.0 mL). The mixture was stirred for 20 min and **7a** (65.2 mg, 0.2 mmol, 1.0 equiv., 98% ee) was then added slowly. The reaction system was stirred for 8 h at rt. After completion of the reaction (TLC), the resulting mixture was purified by flash column chromatography on silica gel (PE/EA 2:1) to afford **7d** (64.7 mg, 99% yield, 98% ee) as white solid.

4.6. Generic Procedure for Iodocyclization Reaction With NIS

To a flame-dried 10 mL sealing tube, **7a** (65.2 mg, 0.2 mmol, 1.0 equiv., 98% ee) and NIS (134.9 mg, 0.6 mmol, 3.0 equiv.) was dissolved in MeCN (2.0 mL). The mixture was cooled to -10 °C and stirred for 48 h. After completion of the reaction (TLC), the resulting mixture was purified by flash column chromatography on silica gel (PE/EA 10:1) to afford **7e** (77.5 mg, 86% yield, 95% ee) as colorless oil.

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Conflict of Interests

The authors declare no conflict of interest

Author Contributions

F.-D.L., L.-Q.L.,, and W.-J.X. directed the investigations and modified the manuscript with revisions provided by the other authors. J.-C.H. and Z.Z. took charge of the DFT calculation. F.-D. L. designed and developed the catalytic method. R.-R.Z. completed a part of the experiment. All the authors were involved the analysis of results and discussions of the project.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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